

Regression Techniques

Problem Sets 1 and 2

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Conventions and important notes

Throughout, vectors are column vectors unless stated otherwise. In Problems Set 1, the notation $\tilde{X}$ in the PDF is written as $\mathbf{X}$ here, and similarly for other random vectors. All calculations are based only on the uploaded assignment and lecture notes.

Important correction in PS1, Q1(3). The printed formula in the uploaded assignment contains a factor that is inconsistent with the direct expansion of the quadratic form. The derivation below gives the correct identity

$$\text{Var}(\mathbf{X}^T A \mathbf{X}) = 2\mu_2^2 \text{tr}(A^2) + (\mu_4 - 3\mu_2^2) \sum_i a_{ii}^2$$

for symmetric A . The document explicitly points out the discrepancy instead of hiding it.

1 Problems Set 1: Random Vectors, Differentiation, Transformations and Multivariate Normal Distribution

1.1 Q1: Random-vector identities

Assignment question

1. Show the following equalities:

- (a) $\text{Var}(\mathbf{Z} - \boldsymbol{\beta}) = \text{Var}(\mathbf{Z})$.
- (b) $\text{Cov}(\mathbf{a}^T \mathbf{Z}, \mathbf{b}^T \mathbf{W}) = \mathbf{a}^T \text{Cov}(\mathbf{Z}, \mathbf{W}) \mathbf{b}$.
- (c) $\text{Cov}(\mathbf{X}_1 + \mathbf{X}_2, \mathbf{Z}_1 + \mathbf{Z}_2) = \text{Cov}(\mathbf{X}_1, \mathbf{Z}_1) + \text{Cov}(\mathbf{X}_1, \mathbf{Z}_2) + \text{Cov}(\mathbf{X}_2, \mathbf{Z}_1) + \text{Cov}(\mathbf{X}_2, \mathbf{Z}_2)$.
- (d) $\text{Var}(\mathbf{X} \pm \mathbf{Z}) = \text{Var}(\mathbf{X}) + \text{Var}(\mathbf{Z}) \pm \text{Cov}(\mathbf{X}, \mathbf{Z}) \pm \text{Cov}(\mathbf{Z}, \mathbf{X})$.
- (e) $\mathbb{E}(\mathbf{Z}^T A \mathbf{Z}) = \boldsymbol{\mu}^T A \boldsymbol{\mu} + \text{tr}(A \Sigma)$, where $\mathbb{E}(\mathbf{Z}) = \boldsymbol{\mu}$ and $\text{Var}(\mathbf{Z}) = \Sigma$.

Required theory (from the uploaded lecture notes)

Random-vector expectation and covariance. For a random vector $\mathbf{Z}$, $\mathbb{E}(\mathbf{Z})$ is the vector of component-wise expectations and

$$\text{Var}(\mathbf{Z}) = \mathbb{E}[(\mathbf{Z} - \boldsymbol{\mu})(\mathbf{Z} - \boldsymbol{\mu})^T], \quad \boldsymbol{\mu} = \mathbb{E}(\mathbf{Z}).$$

For two random vectors, $\text{Cov}(\mathbf{Z}, \mathbf{W})$ is formed component-wise. Constant shifts do not alter covariance. For constant vectors/matrices, covariance is linear in each argument.

For a quadratic form, write $\mathbf{Z} = \boldsymbol{\mu} + \mathbf{U}$, with $\mathbb{E}(\mathbf{U}) = 0$ and $\text{Var}(\mathbf{U}) = \Sigma$. Then

$$\mathbb{E}(\mathbf{Z}^T A \mathbf{Z}) = \boldsymbol{\mu}^T A \boldsymbol{\mu} + \text{tr}(A \Sigma).$$

The trace identity used is $\mathbb{E}(\mathbf{U}^T A \mathbf{U}) = \text{tr}(A \mathbb{E}[\mathbf{U} \mathbf{U}^T]) = \text{tr}(A \Sigma)$.

Part (a): remove the constant. Because $\boldsymbol{\beta}$ is non-random, $\mathbb{E}(\mathbf{Z} - \boldsymbol{\beta}) = \boldsymbol{\mu} - \boldsymbol{\beta}$, while the centered random part is $\mathbf{Z} - \boldsymbol{\mu}$. Therefore

$$\begin{aligned} \text{Var}(\mathbf{Z} - \boldsymbol{\beta}) &= \mathbb{E} \left[((\mathbf{Z} - \boldsymbol{\beta}) - (\boldsymbol{\mu} - \boldsymbol{\beta})) ((\mathbf{Z} - \boldsymbol{\beta}) - (\boldsymbol{\mu} - \boldsymbol{\beta}))^T \right] \\ &= \mathbb{E}[(\mathbf{Z} - \boldsymbol{\mu})(\mathbf{Z} - \boldsymbol{\mu})^T] \\ &= \text{Var}(\mathbf{Z}). \end{aligned}$$

Why this step leads to the next one

Subtracting a fixed vector changes the location of the distribution but not its spread. That is why the covariance matrix is unchanged.

Part (b): covariance of linear forms. Use the definition and pull deterministic row/column vectors outside expectation.

$$\begin{aligned} \text{Cov}(\mathbf{a}^T \mathbf{Z}, \mathbf{b}^T \mathbf{W}) &= \mathbb{E}[(\mathbf{a}^T (\mathbf{Z} - \boldsymbol{\mu}_Z))(\mathbf{b}^T (\mathbf{W} - \boldsymbol{\mu}_W))] \\ &= \mathbf{a}^T \mathbb{E}[(\mathbf{Z} - \boldsymbol{\mu}_Z)(\mathbf{W} - \boldsymbol{\mu}_W)^T] \mathbf{b} \\ &= \mathbf{a}^T \text{Cov}(\mathbf{Z}, \mathbf{W}) \mathbf{b}. \end{aligned}$$

Part (c): expand both sums. Covariance is bilinear, so expand the first argument into two terms and then the second argument into two terms.

$$\begin{aligned}\text{Cov}(\mathbf{X}_1 + \mathbf{X}_2, \mathbf{Z}_1 + \mathbf{Z}_2) &= \text{Cov}(\mathbf{X}_1, \mathbf{Z}_1 + \mathbf{Z}_2) + \text{Cov}(\mathbf{X}_2, \mathbf{Z}_1 + \mathbf{Z}_2) \\ &= \text{Cov}(\mathbf{X}_1, \mathbf{Z}_1) + \text{Cov}(\mathbf{X}_1, \mathbf{Z}_2) + \text{Cov}(\mathbf{X}_2, \mathbf{Z}_1) + \text{Cov}(\mathbf{X}_2, \mathbf{Z}_2).\end{aligned}$$

Part (d): variance is covariance with itself. Use $\text{Var}(\mathbf{A}) = \text{Cov}(\mathbf{A}, \mathbf{A})$ and expand according to part (c).

$$\begin{aligned}\text{Var}(\mathbf{X} \pm \mathbf{Z}) &= \text{Cov}(\mathbf{X} \pm \mathbf{Z}, \mathbf{X} \pm \mathbf{Z}) \\ &= \text{Var}(\mathbf{X}) + \text{Var}(\mathbf{Z}) \pm \text{Cov}(\mathbf{X}, \mathbf{Z}) \pm \text{Cov}(\mathbf{Z}, \mathbf{X}).\end{aligned}$$

Part (e): quadratic form expectation. Set $\mathbf{U} = \mathbf{Z} - \boldsymbol{\mu}$, so $\mathbf{Z} = \boldsymbol{\mu} + \mathbf{U}$, $\mathbb{E}\mathbf{U} = 0$, and $\mathbb{E}(\mathbf{U}\mathbf{U}^T) = \Sigma$. Then

$$\begin{aligned}\mathbf{Z}^T \mathbf{A} \mathbf{Z} &= (\boldsymbol{\mu} + \mathbf{U})^T \mathbf{A} (\boldsymbol{\mu} + \mathbf{U}) \\ &= \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} + 2\boldsymbol{\mu}^T \mathbf{A} \mathbf{U} + \mathbf{U}^T \mathbf{A} \mathbf{U}.\end{aligned}$$

Taking expectations, the middle term vanishes and

$$\mathbb{E}(\mathbf{U}^T \mathbf{A} \mathbf{U}) = \mathbb{E} \text{tr}(\mathbf{A} \mathbf{U} \mathbf{U}^T) = \text{tr}(\mathbf{A} \Sigma).$$

Hence

$$\boxed{\mathbb{E}(\mathbf{Z}^T \mathbf{A} \mathbf{Z}) = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} + \text{tr}(\mathbf{A} \Sigma).}$$

Final answer

All five identities follow directly from the definitions of vector covariance and variance, linearity of expectation, and the trace representation of a quadratic form.

1.2 Q2: Expectation of a matrix product

Assignment question

Let $X = ((X_{ij}))$ be an $m \times n$ matrix. Define $\mathbb{E}(X) = ((\mathbb{E}(X_{ij})))$. Let $A = ((a_{ij}))$ and $B = ((b_{ij}))$ be non-random matrices. Then show that $\mathbb{E}(AXB) = A\mathbb{E}(X)B$.

Required theory (from the uploaded lecture notes)

Matrix expectation. Expectation of a random matrix is defined entry-by-entry. If A and B are deterministic, every entry of AXB is a finite linear combination of the entries of X with deterministic coefficients. Linearity of scalar expectation can therefore be applied entry-by-entry.

Write one generic entry. For the (i, j) entry of AXB ,

$$(AXB)_{ij} = \sum_{r=1}^n \sum_{s=1}^n a_{ir} X_{rs} b_{sj}.$$

Therefore, by linearity of expectation,

$$\begin{aligned}\mathbb{E}[(AXB)_{ij}] &= \sum_{r,s} a_{ir} \mathbb{E}(X_{rs}) b_{sj} \\ &= (A\mathbb{E}(X)B)_{ij}.\end{aligned}$$

Why this step leads to the next one

The two matrices are equal if every corresponding entry is equal. Since the equality above holds for arbitrary i, j , the matrix equality follows.

Final answer

$$\mathbb{E}(AXB) = A\mathbb{E}(X)B.$$

1.3 Q3: Variance of a quadratic form with non-normal independent components**Assignment question**

Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}(X_i) = 0$, $\text{Var}(X_i) = \mu_2$, $\mathbb{E}(X_i^3) = 0$, and $\mathbb{E}(X_i^4) = \mu_4$ for all i . Let $A = ((a_{ij}))$ be a non-random symmetric matrix. Show the stated expression for $\text{Var}(\mathbf{X}^T A \mathbf{X})$.

Required theory (from the uploaded lecture notes)

Quadratic-form variance theory. For independent centered variables with common second moment μ_2 and fourth moment μ_4 , and a symmetric matrix A , the direct expansion gives

$$\text{Var}(\mathbf{X}^T A \mathbf{X}) = 2\mu_2^2 \text{tr}(A^2) + (\mu_4 - 3\mu_2^2) \sum_i a_{ii}^2.$$

Since A is symmetric, $\text{tr}(A^2) = \sum_{i,j} a_{ij}^2$. Equivalently,

$$\text{Var}(\mathbf{X}^T A \mathbf{X}) = \mu_2^2 \left(2 \sum_{i,j} a_{ij}^2 - 3 \sum_i a_{ii}^2 \right) + \mu_4 \sum_i a_{ii}^2.$$

The proof uses the fact that any expectation containing an odd power of one independent centered variable is zero, while terms whose indices pair up survive.

Important note about the PDF. The uploaded PS1 PDF prints a factor $\mu_2^2/2$ in front of the first bracket. The expansion below shows that the mathematically correct coefficient is μ_2^2 (equivalently $2\mu_2^2 \text{tr}(A^2)$ before collecting terms).

Step 1: expand the quadratic form. Because A is symmetric, split the diagonal and off-diagonal parts:

$$Q = \mathbf{X}^T A \mathbf{X} = \sum_i a_{ii} X_i^2 + 2 \sum_{i < j} a_{ij} X_i X_j.$$

Step 2: compute the mean. All off-diagonal terms have expectation $\mathbb{E}(X_i X_j) = \mathbb{E}X_i \mathbb{E}X_j = 0$. Hence

$$\mathbb{E}Q = \mu_2 \sum_i a_{ii} = \mu_2 \text{tr}(A).$$

Step 3: compute the second moment. Square Q . Only terms whose indices pair survive after expectation. There are three types: X_i^4 , $X_i^2 X_j^2$ with $i \neq j$, and terms containing an odd power of some variable, which vanish.

$$\mathbb{E}(Q^2) = \mu_4 \sum_i a_{ii}^2 + 2\mu_2^2 \sum_{i \neq j} a_{ij}^2 + \mu_2^2 \left(\sum_i a_{ii} \right)^2.$$

For a symmetric matrix, $\sum_{i \neq j} a_{ij}^2 = \sum_{i,j} a_{ij}^2 - \sum_i a_{ii}^2$.

Step 4: subtract the square of the mean. Since $(\mathbb{E}Q)^2 = \mu_2^2(\sum_i a_{ii})^2$, the last term cancels, giving

$$\begin{aligned}\text{Var}(Q) &= \mu_4 \sum_i a_{ii}^2 + 2\mu_2^2 \sum_{i \neq j} a_{ij}^2 \\ &= \mu_4 \sum_i a_{ii}^2 + 2\mu_2^2 \left(\sum_{i,j} a_{ij}^2 - \sum_i a_{ii}^2 \right) \\ &= 2\mu_2^2 \sum_{i,j} a_{ij}^2 + (\mu_4 - 2\mu_2^2) \sum_i a_{ii}^2.\end{aligned}$$

However, the diagonal contribution has to be separated once more because $\sum_{i,j} a_{ij}^2$ itself contains the diagonal terms. The standard collected form is therefore

$$\text{Var}(Q) = 2\mu_2^2 \sum_{i,j} a_{ij}^2 + (\mu_4 - 3\mu_2^2) \sum_i a_{ii}^2.$$

Why this step leads to the next one

The key idea is that independence kills every mixed fourth-order term except those in which indices occur in pairs. The fourth moment appears only through the diagonal terms X_i^4 .

Final answer

$$\text{Var}(\mathbf{X}^T \mathbf{A} \mathbf{X}) = 2\mu_2^2 \text{tr}(\mathbf{A}^2) + (\mu_4 - 3\mu_2^2) \sum_i a_{ii}^2.$$

1.4 Differentiation Q1(a): $\|y - Ax\|^2$

Assignment question

Find the maxima and/or minima of

$$g(\mathbf{x}) = \|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2,$$

where A is a positive definite invertible matrix.

Required theory (from the uploaded lecture notes)

Gradient/Hessian test. For $g : \mathbb{R}^p \rightarrow \mathbb{R}$, an interior stationary point satisfies $\nabla g = 0$. A critical point is a local minimum when the Hessian is positive definite and a local maximum when it is negative definite. For symmetric A ,

$$\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{A} \mathbf{x}) = 2\mathbf{A}\mathbf{x}, \quad \nabla_{\mathbf{x}}(\mathbf{a}^T \mathbf{x}) = \mathbf{a},$$

and the Hessian of the quadratic is constant.

Step 1: rewrite the objective. Use $\|\mathbf{v}\|^2 = \mathbf{v}^T \mathbf{v}$ to obtain

$$\begin{aligned}g(\mathbf{x}) &= (\mathbf{y} - \mathbf{A}\mathbf{x})^T (\mathbf{y} - \mathbf{A}\mathbf{x}) \\ &= \mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T \mathbf{A}\mathbf{x} + \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x}.\end{aligned}$$

Because positive definite matrices are symmetric, $A^T = A$, so $A^T A = A^2$.)

Step 2: calculate the gradient. Differentiate with respect to $\mathbf{x}$:

$$\nabla g(\mathbf{x}) = -2A^T \mathbf{y} + 2A^T A\mathbf{x} = 2A(A\mathbf{x} - \mathbf{y}).$$

Step 3: solve the first-order condition. Since A is invertible, $A(A\mathbf{x} - \mathbf{y}) = 0$ implies

$$A\mathbf{x} = \mathbf{y} \implies \boxed{\mathbf{x}^* = A^{-1}\mathbf{y}}.$$

Step 4: classify the stationary point. The Hessian is

$$H = 2A^T A = 2A^2.$$

If A is positive definite, then A^2 is positive definite, hence H is positive definite. Therefore $\mathbf{x}^*$ is a strict local minimum. Since $g(\mathbf{x}) \rightarrow \infty$ as $\|\mathbf{x}\| \rightarrow \infty$, this is also the unique global minimum. At the minimizer $A\mathbf{x}^* = \mathbf{y}$, so $g(\mathbf{x}^*) = 0$.

Final answer

$$\boxed{\mathbf{x}^* = A^{-1}\mathbf{y} \text{ is the unique global minimum, with } g(\mathbf{x}^*) = 0.}$$

There is no global maximum because the quadratic grows without bound.

1.5 Differentiation Q1(b): $\sum_j (x_j^4 - 4x_j^2)$

Assignment question

Find the maxima and/or minima of

$$g(\mathbf{x}) = \sum_{j=1}^k (x_j^4 - 4x_j^2).$$

Required theory (from the uploaded lecture notes)

Stationary-point and Hessian rule. Solve $\nabla g = 0$. Then use the Hessian. A positive definite Hessian gives a local minimum; a negative definite Hessian gives a local maximum; an indefinite Hessian gives a saddle point.

Step 1: gradient. For each coordinate, $\partial(x_j^4 - 4x_j^2)/\partial x_j = 4x_j^3 - 8x_j = 4x_j(x_j^2 - 2)$. Hence

$$\nabla g(\mathbf{x}) = \begin{pmatrix} 4x_1(x_1^2 - 2) \\ \vdots \\ 4x_k(x_k^2 - 2) \end{pmatrix}.$$

Thus every coordinate satisfies $x_j \in \{0, \sqrt{2}, -\sqrt{2}\}$. There are 3^k stationary points.

Step 2: Hessian. Because the coordinates separate, the Hessian is diagonal:

$$H(\mathbf{x}) = \text{diag}(12x_1^2 - 8, \dots, 12x_k^2 - 8).$$

Step 3: classify. At $\mathbf{0}$, $H = -8I$, so $\mathbf{0}$ is a strict local maximum. If every coordinate is $\pm\sqrt{2}$, every diagonal entry is 16, so $H = 16I$ and the point is a strict local minimum. If at least one coordinate is 0 and at least one is $\pm\sqrt{2}$, the Hessian has both negative and positive diagonal entries and is indefinite: the point is a saddle point.

Step 4: decide whether the extrema are global. For one coordinate, $x^4 - 4x^2 \rightarrow \infty$ as $|x| \rightarrow \infty$. Therefore there is no global maximum. At $x = \pm\sqrt{2}$, the one-coordinate value is $4 - 8 = -4$, so the minimum of the sum is $-4k$.

Final answer

$\mathbf{0}$ is a strict local maximum;

$x_j \in \{\pm\sqrt{2}\} \forall j$ are strict global minima with $g = -4k$;

all remaining critical points are saddles, and there is no global maximum.

1.6 Differentiation Q1(c): $\|x\|^2 + e^{a^T x}$

Assignment question

Find the maxima and/or minima of

$$g(\mathbf{x}) = \|\mathbf{x}\|^2 + \exp(\mathbf{a}^T \mathbf{x}).$$

Required theory (from the uploaded lecture notes)

For $g : \mathbb{R}^p \rightarrow \mathbb{R}$, solve $\nabla g = 0$ and classify with the Hessian. Also, $\|\mathbf{x}\|^2$ is coercive and $e^{\mathbf{a}^T \mathbf{x}} > 0$, so a global minimizer is expected. The Hessian may be tested through $\mathbf{v}^T H \mathbf{v}$ for arbitrary nonzero $\mathbf{v}$.

Step 1: gradient. We have $\text{grad}\|\mathbf{x}\|^2 = 2\mathbf{x}$ and $\text{grad}e^{\mathbf{a}^T \mathbf{x}} = e^{\mathbf{a}^T \mathbf{x}}\mathbf{a}$, hence

$$\nabla g(\mathbf{x}) = 2\mathbf{x} + e^{\mathbf{a}^T \mathbf{x}}\mathbf{a}.$$

Step 2: stationary equation. Set the gradient to zero: $2\mathbf{x} = -e^{\mathbf{a}^T \mathbf{x}}\mathbf{a}$. Therefore the stationary point must lie in the span of $\mathbf{a}$; write $\mathbf{x} = t\mathbf{a}$. Then

$$2t + e^{t\|\mathbf{a}\|^2} = 0.$$

Let $s = t\|\mathbf{a}\|^2$. Then se^{-s} is best handled through Lambert W : from $2t = -e^{t\|\mathbf{a}\|^2}$, with $c = \|a\|^2/2$, one gets

$$(-t\|\mathbf{a}\|^2)e^{-t\|\mathbf{a}\|^2} = c.$$

Hence $-t\|\mathbf{a}\|^2 = W(c)$ and

$$\mathbf{x}^* = -\frac{W(\|a\|^2/2)}{\|a\|^2}\mathbf{a}$$

for $\mathbf{a} \neq 0$. If $\mathbf{a} = 0$, clearly $\mathbf{x}^* = 0$.

Step 3: Hessian. Differentiate the gradient once more:

$$H(\mathbf{x}) = 2I + e^{\mathbf{a}^T \mathbf{x}}\mathbf{a}\mathbf{a}^T.$$

For any nonzero $\mathbf{v}$,

$$\mathbf{v}^T H(\mathbf{x}) \mathbf{v} = 2\|\mathbf{v}\|^2 + e^{\mathbf{a}^T \mathbf{x}}(\mathbf{a}^T \mathbf{v})^2 > 0.$$

Thus H is positive definite everywhere. Therefore g is strictly convex and the unique stationary point is the unique global minimum.

Why this step leads to the next one

There cannot be a maximum because $g(\mathbf{x}) \geq \|\mathbf{x}\|^2 \rightarrow \infty$ as $\|\mathbf{x}\| \rightarrow \infty$.

Final answer

$$\mathbf{x}^* = -\frac{W(\|a\|^2/2)}{\|a\|^2}\mathbf{a} \text{ is the unique global minimum;}$$

there is no maximum.

1.7 Transformation Q2: $U = X + Y$, $V = X - Y$

Assignment question

Let (X, Y) have joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define $U = X + Y$, $V = X - Y$. Find the joint distribution of (U, V) .

Required theory (from the uploaded lecture notes)

Multivariate change of variables. If $\mathbf{Z} = h(\mathbf{X})$ is one-to-one, then

$$f_{\mathbf{Z}}(\mathbf{z}) = f_{\mathbf{X}}(h^{-1}(\mathbf{z})) \left| \det \frac{\partial \mathbf{x}}{\partial \mathbf{z}} \right|.$$

The support must also be transformed exactly; the Jacobian alone is not enough.

Step 1: invert the transformation. Solve $u = x + y$ and $v = x - y$:

$$x = \frac{u+v}{2}, \quad y = \frac{u-v}{2}.$$

Step 2: calculate the Jacobian. Therefore

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix} = -\frac{1}{2}, \quad |J| = \frac{1}{2}.$$

Step 3: transform the support. From $0 < y < x < 1$ we get $v > 0$, $u > v$, and $u + v < 2$. Equivalently,

$$0 < u < 2, \quad 0 < v < \min(u, 2 - u).$$

Step 4: transform the density. Since the original density is 2 on the support,

$$f_{U,V}(u, v) = 2 \cdot \frac{1}{2} = 1$$

on the transformed support and 0 elsewhere.

Final answer

$$f_{U,V}(u, v) = \begin{cases} 1, & 0 < u < 2, \ 0 < v < \min(u, 2 - u), \\ 0, & \text{otherwise.} \end{cases}$$

1.8 Transformation Q3: $U = XY, V = Y$

Assignment question

Let (X, Y) have joint pdf

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Define $U = XY, V = Y$. Find the joint distribution of (U, V) .

Required theory (from the uploaded lecture notes)

Use the same one-to-one change-of-variables formula. Always compute (i) inverse transformation, (ii) absolute Jacobian, and (iii) transformed support.

Step 1: inverse. Since $v = y$ and $u = xy$,

$$y = v, \quad x = \frac{u}{v}.$$

The inverse requires $v > 0$.

Step 2: Jacobian. We have

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1/v & -u/v^2 \\ 0 & 1 \end{vmatrix} = \frac{1}{v}.$$

Step 3: support. The original conditions imply $0 < v < 1$ and $0 < u/v < 1$, so $0 < u < v < 1$.

Step 4: density. Substitute $x = u/v, y = v$ and multiply by $1/v$:

$$f_{U,V}(u, v) = 4 \frac{u}{v} \frac{1}{v} = \frac{4u}{v}.$$

Final answer

$$f_{U,V}(u, v) = \begin{cases} \frac{4u}{v}, & 0 < u < v < 1, \\ 0, & \text{otherwise.} \end{cases}$$

1.9 Transformation Q4: $U = X, V = Y/X$

Assignment question

Let (X, Y) have joint pdf

$$f_{X,Y}(x, y) = \begin{cases} e^{-(x+y)}, & x > 0, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Define $U = X, V = Y/X$. Find the joint distribution of (U, V) .

Required theory (from the uploaded lecture notes)

For a one-to-one transformation, $f_{U,V} = f_{X,Y}(x(u, v), y(u, v))|J|$. The support is obtained by translating all original inequalities into u, v .

Step 1: inverse. We have $u = x$ and $v = y/x$, hence $x = u$, $y = uv$.

Step 2: Jacobian. Therefore

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} 1 & 0 \\ v & u \end{vmatrix} = u, \quad |J| = u.$$

Step 3: support. The conditions $x > 0$ and $y > 0$ become $u > 0$ and $uv > 0$. Thus $u > 0, v > 0$.

Step 4: density. Substitution gives

$$f_{U,V}(u, v) = e^{-u-uv}u, \quad u > 0, v > 0.$$

Final answer

$$f_{U,V}(u, v) = \begin{cases} ue^{-u(1+v)}, & u > 0, v > 0, \\ 0, & \text{otherwise.} \end{cases}$$

1.10 Transformation Q5: ratios $U_1 = X_1$, $U_i = X_i/X_1$

Assignment question

Let $(X_1, \dots, X_k)$ have joint pdf

$$f_{X_1, \dots, X_k}(x_1, \dots, x_k) = \begin{cases} e^{-(x_1 + \dots + x_k)}, & x_i > 0, i = 1, \dots, k, \\ 0, & \text{otherwise.} \end{cases}$$

Define $U_1 = X_1$ and $U_i = X_i/X_1$ for $i = 2, \dots, k$. Find the joint distribution of $\{U_1, \dots, U_k\}$.

Required theory (from the uploaded lecture notes)

Use the multivariate Jacobian formula. For triangular transformations of the form $x_1 = u_1$ and $x_i = u_1 u_i$, the Jacobian is triangular and its determinant is easy to compute. Support must remain positive.

Step 1: inverse. From the definitions,

$$x_1 = u_1, \quad x_i = u_1 u_i, \quad i = 2, \dots, k.$$

Step 2: Jacobian. The Jacobian matrix is triangular after ordering variables as $(u_1, u_2, \dots, u_k)$, with diagonal entries $1, u_1, \dots, u_1$. Hence

$$\left| \frac{\partial(x_1, \dots, x_k)}{\partial(u_1, \dots, u_k)} \right| = u_1^{k-1}.$$

Step 3: support. All $x_i > 0$ imply $u_1 > 0$ and $u_i > 0$ for $i = 2, \dots, k$.

Step 4: density. The exponent becomes

$$x_1 + \dots + x_k = u_1 \left(1 + \sum_{i=2}^k u_i \right).$$

Therefore

$$f_{U_1, \dots, U_k}(u_1, \dots, u_k) = u_1^{k-1} \exp \left[-u_1 \left(1 + \sum_{i=2}^k u_i \right) \right], \quad u_i > 0.$$

Why this step leads to the next one

The factor u_1^{k-1} comes entirely from the Jacobian, while the exponential part comes from substituting the inverse transformation into the original density.

1.11 MVN Q1: $Y = \text{sign}(X)$ and $Z = |X|$

Assignment question

Let $X \sim N(0, 1)$. Define $Y = \text{sign}(X)$ with $Y = 1$ for $X \geq 0$ and $Y = -1$ otherwise, and $Z = |X|$.

- (a) Find the marginal distributions of Y and Z .
- (b) Find the joint CDF of (Y, Z) . Hence or otherwise prove that Y and Z are independently distributed.

Required theory (from the uploaded lecture notes)

Normal distribution facts. If $X \sim N(0, 1)$, symmetry gives $\mathbb{P}(X \geq 0) = \mathbb{P}(X < 0) = 1/2$. The absolute value has the half-normal density

$$f_Z(z) = f_X(z) + f_X(-z) = 2\varphi(z) = \sqrt{\frac{2}{\pi}} e^{-z^2/2}, \quad z \geq 0.$$

Independence can be established by showing that the joint CDF factors into the product of the marginal CDFs.

Part (a): marginals. For Y , $\mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = 1/2$. For Z , for $z \geq 0$,

$$F_Z(z) = \mathbb{P}(|X| \leq z) = \mathbb{P}(-z \leq X \leq z) = 2\Phi(z) - 1,$$

and thus $f_Z(z) = 2\phi(z)$ for $z > 0$.

Part (b): joint CDF. Because Y is discrete, write the CDF piecewise. For $z < 0$, it is 0. For $z \geq 0$,

$$F_{Y,Z}(y, z) = \begin{cases} 0, & y < -1, \\ \Phi(z) - \frac{1}{2}, & -1 \leq y < 1, \\ 2\Phi(z) - 1, & y \geq 1. \end{cases}$$

Step 3: factorization. For $-1 \leq y < 1$, the marginal CDF of Y is $1/2$ and the marginal CDF of Z is $2\Phi(z) - 1$, so

$$F_{Y,Z}(y, z) = \frac{1}{2} (2\Phi(z) - 1) = F_Y(y) F_Z(z).$$

The other ranges also match the product. Therefore Y and Z are independent.

Final answer

$$Y \sim \text{Rademacher}(\pm 1), \quad \mathbb{P}(Y = 1) = \mathbb{P}(Y = -1) = \frac{1}{2},$$

$$f_Z(z) = \sqrt{\frac{2}{\pi}} e^{-z^2/2}, \quad z \geq 0, \quad Y \perp Z.$$

1.12 MVN Q2: mixture with opposite correlations

Assignment question

Let the joint distribution of $\mathbf{X}$ be an equal mixture of $N_4(\mathbf{0}, \Sigma_1)$ and $N_4(\mathbf{0}, \Sigma_2)$, where

$$\Sigma_1 = \begin{pmatrix} E_{2,\rho} & 0 \\ 0 & E_{2,\rho} \end{pmatrix}, \quad \Sigma_2 = \begin{pmatrix} E_{2,\rho} & 0 \\ 0 & E_{2,-\rho} \end{pmatrix},$$

and $E_{2,\rho} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, $-1 < \rho < 1$. (a) Find the joint pdfs of (X_1, X_2) and (X_3, X_4) . (b) Prove that X_4 and X_3 are uncorrelated but not independent. (c) Are (X_1, X_2) and (X_3, X_4) independent?

Required theory (from the uploaded lecture notes)

Mixture and bivariate normal facts. A 2-variate standard normal pair with correlation ρ has density

$$\phi_\rho(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{x_1^2 - 2\rho x_1 x_2 + x_2^2}{2(1-\rho^2)}\right].$$

In a jointly normal vector, zero covariance is equivalent to independence. For a non-normal mixture, zero covariance does not imply independence.

Part (a): first pair. Under both mixture components, $(X_1, X_2) \sim N_2(0, E_{2,\rho})$. Therefore

$$f_{X_1, X_2}(x_1, x_2) = \phi_\rho(x_1, x_2).$$

Part (a): second pair. For (X_3, X_4) , the two mixture components have correlations ρ and $-\rho$, respectively. Hence

$$f_{X_3, X_4}(x_3, x_4) = \frac{1}{2}\phi_\rho(x_3, x_4) + \frac{1}{2}\phi_{-\rho}(x_3, x_4).$$

Part (b): covariance. Both marginal variances equal 1. The two mixture components contribute covariances ρ and $-\rho$, so

$$\text{Cov}(X_3, X_4) = \frac{1}{2}\rho + \frac{1}{2}(-\rho) = 0.$$

To prove dependence for $\rho \neq 0$, compare the mixed fourth moment. For a standard bivariate normal pair, $\mathbb{E}(X_3^2 X_4^2) = 1 + 2\rho^2$ regardless of the sign of ρ , so the mixture also has

$$\mathbb{E}(X_3^2 X_4^2) = 1 + 2\rho^2 > 1 = \mathbb{E}(X_3^2)\mathbb{E}(X_4^2)$$

when $\rho \neq 0$. Hence the variables are not independent. At the exceptional case $\rho = 0$, the two are independent, so the statement “not independent” requires $\rho \neq 0$.

Part (c): block independence. Although the second block is a mixture, the first block has the same distribution under both mixture components. Therefore the component label does not alter the distribution of (X_1, X_2) . Algebraically, if f_{12} is the common density of the first block and $f_{34}^{(+)}$, $f_{34}^{(-)}$ are the two conditional densities of the second block, then

$$f_{12,34} = \frac{1}{2}f_{12}f_{34}^{(+)} + \frac{1}{2}f_{12}f_{34}^{(-)} = f_{12}\left(\frac{1}{2}f_{34}^{(+)} + \frac{1}{2}f_{34}^{(-)}\right) = f_{12}f_{34}.$$

Final answer

$$(X_1, X_2) \perp\!\!\!\perp (X_3, X_4) \text{ for every } -1 < \rho < 1.$$

1.13 MVN Q3: normal marginals and non-normal joint vector

Assignment question

Let $\mathbf{X} = (X_1, X_2, X_3)^T$ have joint pdf

$$f_{\mathbf{X}}(\mathbf{x}) = (2\pi)^{-3/2} e^{-(x_1^2 + x_2^2 + x_3^2)/2} \left\{ 1 + x_1 x_2 x_3 e^{-(x_1^2 + x_2^2 + x_3^2)/2} \right\}.$$

- (a) Show $X_i \sim N(0, 1)$. (b) Show (X_i, X_j) are bivariate normal. Hence find $\mathbb{E}\mathbf{X}$ and $\text{Var}(\mathbf{X})$.
(c) Does $\mathbf{X}$ follow a multivariate normal distribution?

Required theory (from the uploaded lecture notes)

Multivariate normal facts. A vector is multivariate normal if every linear combination is univariate normal. Every marginal of a multivariate normal vector is normal. However, the converse is false: normal marginals and even normal bivariate marginals do not necessarily imply joint multivariate normality. When a joint vector is multivariate normal, uncorrelated components are independent.

Part (a): integrate out two coordinates. For X_1 , integrate over x_2, x_3 . The base Gaussian term integrates to $\phi(x_1)$. The perturbation contains $x_2 x_3$ multiplied by even functions of x_2, x_3 , so each integral is zero because the integrand is odd. Thus

$$f_{X_1}(x_1) = \phi(x_1).$$

The same argument works for X_2 and X_3 , hence $X_i \sim N(0, 1)$.

Part (b): integrate out the remaining coordinate. For (X_1, X_2) , integrate over x_3 . The perturbation term contains $x_3 e^{-x_3^2/2}$, an odd function whose integral over $\mathbb{R}$ is zero. Therefore

$$f_{X_1, X_2}(x_1, x_2) = \phi(x_1)\phi(x_2).$$

Hence every pair (X_i, X_j) is $N_2(0, I_2)$.

Part (b): assemble mean and covariance. Therefore

$$\mathbb{E}\mathbf{X} = \mathbf{0}, \quad \text{Var}(\mathbf{X}) = I_3.$$

Part (c): test joint normality. If $\mathbf{X}$ were $N_3(0, I_3)$, then its three components would have to be mutually independent, because for a multivariate normal vector zero covariance implies independence. But the displayed joint density contains the nonzero interaction term $x_1 x_2 x_3 e^{-r^2/2}$, so it does not factor into $f_{X_1} f_{X_2} f_{X_3}$. Therefore the components are not jointly independent and the vector is not multivariate normal.

Final answer

$\mathbf{X}$ is not multivariate normal despite having normal univariate and bivariate marginals.

1.14 MVN Q4: high-density ellipses

Assignment question

Let $\mathbf{Z} = (X, Y)^T$ be bivariate normal with parameters $(1, 2, 2.5, 1.5, 0.5)$, and let $f_{\mathbf{Z}}$ be its pdf. Identify $\rho(c) = \{\mathbf{z} : f_{\mathbf{Z}}(\mathbf{z}) \geq c\}$. Draw $\rho(0.05)$ and $\rho(0.025)$.

Required theory (from the uploaded lecture notes)

Density and level sets of a bivariate normal. For mean $\boldsymbol{\mu}$ and covariance

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix},$$

the density is

$$f(\mathbf{z}) = \frac{1}{2\pi|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(\mathbf{z} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{z} - \boldsymbol{\mu})\right].$$

Therefore $f(\mathbf{z}) \geq c$ is equivalent to an ellipse

$$(\mathbf{z} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{z} - \boldsymbol{\mu}) \leq -2 \log \left(c(2\pi)|\Sigma|^{1/2} \right).$$

Step 1: identify parameters. Interpret the parameter tuple as $\mu_X = 1$, $\mu_Y = 2$, $\sigma_X^2 = 2.5$, $\sigma_Y^2 = 1.5$, $\rho = 0.5$. Thus

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 2.5 & 0.5\sqrt{2.5}\sqrt{1.5} \\ 0.5\sqrt{2.5}\sqrt{1.5} & 1.5 \end{pmatrix}.$$

Since $0.5\sqrt{2.5}\sqrt{1.5} = \sqrt{15}/4$,

$$\Sigma = \begin{pmatrix} 2.5 & \frac{\sqrt{15}}{4} \\ \frac{\sqrt{15}}{4} & 1.5 \end{pmatrix}, \quad |\Sigma| = 2.5(1.5)(1 - 0.25) = \frac{45}{32}.$$

Step 2: write the quadratic form. Since $\rho = 1/2$,

$$(\mathbf{z} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{z} - \boldsymbol{\mu}) = \frac{1}{1 - \rho^2} \left[\frac{(x - 1)^2}{2.5} - \frac{2\rho(x - 1)(y - 2)}{\sqrt{2.5}\sqrt{1.5}} + \frac{(y - 2)^2}{1.5} \right].$$

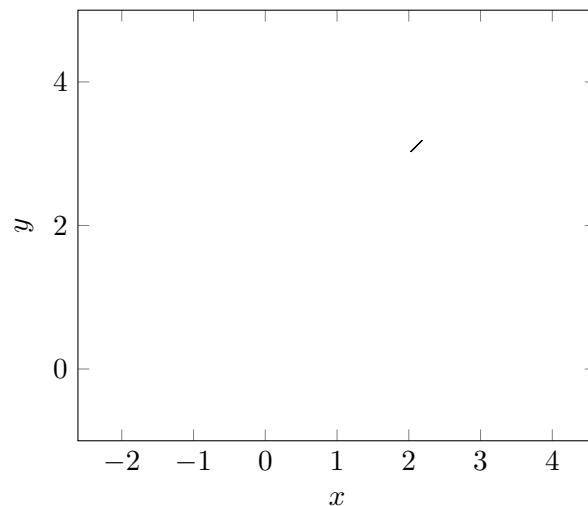
Thus the set $\rho(c)$ is the interior of the ellipse defined by this quadratic being at most q_c below.

Step 3: compute the right-hand sides. The maximum density is $f(\boldsymbol{\mu}) = 1/(2\pi|\Sigma|^{1/2})$. For a density threshold c , define

$$q(c) = -2 \log \left(c(2\pi)|\Sigma|^{1/2} \right).$$

Then $f(\mathbf{z}) \geq c$ iff $(\mathbf{z} - \boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{z} - \boldsymbol{\mu}) \leq q(c)$. For $c = 0.05$ and 0.025 , the values are approximately $q(0.05) = 0.96123$ and $q(0.025) = 2.00095$.

Step 4: plotting code. The following `pgfplots` snippet parameterizes the two ellipses using the eigen-decomposition of Σ .



Final answer

$$\rho(c) = \left\{ \mathbf{z} : (\mathbf{z} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{z} - \boldsymbol{\mu}) \leq -2 \log \left(c(2\pi) |\Sigma|^{1/2} \right) \right\}.$$

The two requested sets are concentric ellipses centered at $(1, 2)$, with $\rho(0.05) \subset \rho(0.025)$.

1.15 MVN Q5: Gaussian random-effects decomposition**Assignment question**

Let $\mathbf{y} = X\boldsymbol{\beta} + Z\mathbf{b} + \boldsymbol{\varepsilon}$, where X, Z are fixed $p \times q$ and $p \times r$ matrices, $\mathbf{b} \sim N_r(0, \Delta)$, $\boldsymbol{\varepsilon} \sim N_p(0, \sigma^2 I_p)$, and $\mathbf{b}$ and $\boldsymbol{\varepsilon}$ are independent. Show: (a) $\mathbf{y} \sim N_p(X\boldsymbol{\beta}, W)$, $W = Z\Delta Z^T + \sigma^2 I_p$; (b) the joint distribution of $(\mathbf{y}, \mathbf{b})^T$; (c) $\mathbb{E}(\mathbf{b} \mid \mathbf{y}) = \Delta Z W^{-1}(\mathbf{y} - X\boldsymbol{\beta})$; (d) $\mathbf{y} \mid \mathbf{b} \sim N_p(X\boldsymbol{\beta} + Z\mathbf{b}, \sigma^2 I_p)$.

Required theory (from the uploaded lecture notes)

Normal linear transformation and conditional normal distribution. If $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$ and $\mathbf{Y} = A\mathbf{X} + b$, then $\mathbf{Y} \sim N(A\boldsymbol{\mu} + b, A\Sigma A^T)$. For a jointly normal partition

$$\begin{pmatrix} \mathbf{U}_1 \\ \mathbf{U}_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right),$$

the conditional distribution is

$$\mathbf{U}_1 \mid \mathbf{U}_2 = u_2 \sim N(\boldsymbol{\mu}_1 + \Sigma_{12} \Sigma_{22}^{-1} (u_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}).$$

Part (a): distribution of $\mathbf{y}$. Because $\mathbf{y}$ is an affine transformation of the jointly normal vector $(\mathbf{b}^T, \boldsymbol{\varepsilon}^T)^T$, it is normal. Its mean is

$$\mathbb{E}\mathbf{y} = X\boldsymbol{\beta} + Z\mathbb{E}\mathbf{b} + \mathbb{E}\boldsymbol{\varepsilon} = X\boldsymbol{\beta}.$$

Independence gives $\text{Cov}(\mathbf{b}, \boldsymbol{\varepsilon}) = 0$, so

$$\text{Var}(\mathbf{y}) = Z\Delta Z^T + \sigma^2 I_p =: W.$$

Therefore $\mathbf{y} \sim N_p(X\boldsymbol{\beta}, W)$.

Part (b): joint distribution. Write

$$\begin{pmatrix} \mathbf{y} \\ \mathbf{b} \end{pmatrix} = \begin{pmatrix} X\boldsymbol{\beta} \\ 0 \end{pmatrix} + \begin{pmatrix} Z & I_p \\ I_r & 0 \end{pmatrix} \begin{pmatrix} \mathbf{b} \\ \boldsymbol{\varepsilon} \end{pmatrix} \text{ (appropriately dimensioned independent Gaussian vector).}$$

More directly, compute the blocks. We have $\text{Var}(\mathbf{y}) = W$, $\text{Var}(\mathbf{b}) = \Delta$, and

$$\text{Cov}(\mathbf{y}, \mathbf{b}) = \text{Cov}(Z\mathbf{b} + \boldsymbol{\varepsilon}, \mathbf{b}) = Z\Delta.$$

Hence

$$\begin{pmatrix} \mathbf{y} \\ \mathbf{b} \end{pmatrix} \sim N_{p+r} \left(\begin{pmatrix} X\boldsymbol{\beta} \\ 0 \end{pmatrix}, \begin{pmatrix} W & Z\Delta \\ \Delta Z^T & \Delta \end{pmatrix} \right).$$

Part (c): conditional expectation. Apply the conditional mean formula with $\mathbf{U}_1 = \mathbf{b}$ and $\mathbf{U}_2 = \mathbf{y}$. Thus

$$\mathbb{E}(\mathbf{b} \mid \mathbf{y}) = 0 + \Delta Z^T W^{-1}(\mathbf{y} - X\boldsymbol{\beta}).$$

Part (d): conditional on b . Given $\mathbf{b}$, the only remaining randomness is ε :

$$\mathbf{y} \mid \mathbf{b} = X\beta + Z\mathbf{b} + \varepsilon,$$

so

$$\mathbf{y} \mid \mathbf{b} \sim N_p(X\beta + Z\mathbf{b}, \sigma^2 I_p).$$

Final answer

All four results follow from affine transformations of normal vectors and the standard conditional-normal formula.

1.16 MVN Q6: MGF

Assignment question

Let $\mathbf{Y} \sim N_p(\mu, \Sigma)$ where $\Sigma > 0$. Define

$$M_{\mathbf{Y}}(t) = \mathbb{E} \exp(t^T \mathbf{Y}).$$

Without explicitly deriving, find $M_{\mathbf{Y}}(t)$ using normal-distribution properties and the univariate normal MGF.

Required theory (from the uploaded lecture notes)

If $\mathbf{Y} \sim N_p(\mu, \Sigma)$, every linear form $t^T \mathbf{Y}$ is univariate normal with mean $t^T \mu$ and variance $t^T \Sigma t$. For $V \sim N(m, s^2)$, $M_V(1) = \mathbb{E}(e^V) = e^{m+s^2/2}$.

Apply the linear-form property. Since $t^T \mathbf{Y} \sim N(t^T \mu, t^T \Sigma t)$,

$$M_{\mathbf{Y}}(t) = \exp \left(t^T \mu + \frac{1}{2} t^T \Sigma t \right).$$

Final answer

$$M_{\mathbf{Y}}(t) = \exp \left(t^T \mu + \frac{1}{2} t^T \Sigma t \right).$$

1.17 MVN Q7: orthonormal-row linear transformation

Assignment question

Let $\mathbf{Y} \sim N_p(\mu, I)$ and U be a rectangular matrix with $UU^T = I_q$, $q < p$. Find the distribution of $U\mathbf{Y}$.

Required theory (from the uploaded lecture notes)

A linear transformation of a multivariate normal vector is normal: $A\mathbf{Y} \sim N(A\mu, A\Sigma A^T)$.

Apply the theorem. Here $A = U$ and $\Sigma = I_p$, so

$$U\mathbf{Y} \sim N_q(U\mu, UI_p U^T) = N_q(U\mu, UU^T) = N_q(U\mu, I_q).$$

Final answer

$$U\mathbf{Y} \sim N_q(U\mu, I_q).$$

1.18 MVN Q8: characteristic function and independence of $\bar{Y}$ and residual sum of squares**Assignment question**

Suppose $Y_1, \dots, Y_n$ are independent $N(0, 1)$. Calculate the characteristic function of the random vector

$$(\bar{Y}, Y_1 - \bar{Y}, \dots, Y_n - \bar{Y})$$

and hence show that $\bar{Y}$ is independent of $\sum_{i=1}^n (Y_i - \bar{Y})^2$.

Required theory (from the uploaded lecture notes)

For $Y \sim N(0, I)$ and any deterministic vector c , $c^T Y \sim N(0, c^T c)$, so its characteristic function is $\exp(-\frac{1}{2}c^T c)$. Independence of jointly normal subvectors follows from zero cross-covariance, equivalently factorization of the joint characteristic function.

Step 1: write the generic linear combination. Let t_0 correspond to $\bar{Y}$ and $t_1, \dots, t_n$ to residuals. Put $\bar{t} = n^{-1} \sum_i t_i$. Then

$$\begin{aligned} t_0 \bar{Y} + \sum_{i=1}^n t_i (Y_i - \bar{Y}) \\ &= \sum_{i=1}^n \left[t_i + \frac{t_0 - \sum_j t_j}{n} \right] Y_i \\ &= \sum_{i=1}^n (t_i - \bar{t}) Y_i + \frac{t_0}{n} \sum_{i=1}^n Y_i. \end{aligned}$$

The sum of squared coefficients is

$$\sum_{i=1}^n (t_i - \bar{t})^2 + \frac{t_0^2}{n}.$$

Step 2: characteristic function. Therefore

$$\varphi(t_0, t_1, \dots, t_n) = \exp\left(-\frac{t_0^2}{2n}\right) \exp\left(-\frac{1}{2} \sum_{i=1}^n (t_i - \bar{t})^2\right).$$

This is a product of a function of t_0 and a function of $(t_1, \dots, t_n)$. Hence $\bar{Y}$ is independent of the whole residual vector, and therefore independent of any measurable function of it, in particular $\sum_i (Y_i - \bar{Y})^2$.

Final answer

$$\bar{Y} \perp \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

1.19 MVN Q9: Helmert transformation

Assignment question

Let X_1, X_2, X_3 be i.i.d. $N(0, 1)$ and define

$$Y_1 = \frac{X_1 + X_2 + X_3}{\sqrt{3}}, \quad Y_2 = \frac{X_1 - X_2}{\sqrt{2}}, \quad Y_3 = \frac{X_1 + X_2 - 2X_3}{\sqrt{6}}.$$

Show that Y_1, Y_2, Y_3 are i.i.d. $N(0, 1)$.

Required theory (from the uploaded lecture notes)

If $\mathbf{X} \sim N_p(0, I)$ and A is orthogonal, $AA^T = I$, then $A\mathbf{X} \sim N_p(0, I)$. A $N_p(0, I)$ vector has independent standard-normal components.

Step 1: form the transformation matrix. Write $\mathbf{Y} = A\mathbf{X}$ with rows

$$A = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \end{pmatrix}.$$

Step 2: verify orthonormality. Each row has squared norm 1, and every pair of distinct rows has inner product 0. Hence $AA^T = I_3$.

Step 3: transform the distribution. Since $\mathbf{X} \sim N_3(0, I_3)$,

$$\mathbf{Y} \sim N_3(0, AA^T) = N_3(0, I_3).$$

Thus the components are independent $N(0, 1)$.

Final answer

$$Y_1, Y_2, Y_3 \stackrel{\text{i.i.d.}}{\sim} N(0, 1).$$

1.20 MVN Q10: independence of two linear forms

Assignment question

Given $\mathbf{Y} \sim N_3(\mu, \Sigma)$ with

$$\Sigma = \sigma^2 \begin{pmatrix} 1 & \rho & 0 \\ \rho & 1 & \rho \\ 0 & \rho & 1 \end{pmatrix},$$

for what value(s) of ρ are $Y_1 + Y_2 + Y_3$ and $Y_1 - Y_2 - Y_3$ statistically independent?

Required theory (from the uploaded lecture notes)

For jointly normal random variables, independence is equivalent to zero covariance. For linear forms $a^T Y$ and $b^T Y$, the covariance is $a^T \Sigma b$.

Step 1: choose coefficient vectors. Let $a = (1, 1, 1)^T$ and $b = (1, -1, -1)^T$.

Step 2: calculate covariance. Direct multiplication gives

$$\text{Cov}(a^T Y, b^T Y) = a^T \Sigma b = \sigma^2(-1 - 2\rho).$$

Step 3: set covariance to zero. Thus independence requires $-1 - 2\rho = 0$, so

$$\rho = -\frac{1}{2}.$$

Step 4: check admissibility. For $\rho = -1/2$, the covariance matrix is positive definite (its eigenvalues are positive), so the value is valid.

Final answer

$$\rho = -\frac{1}{2}.$$

1.21 MVN Q11: difference of chi-squares from nested projections

Assignment question

Let $\mathbf{Y} \sim N_n(\theta, \sigma^2 I)$ and

$$Q_i = \frac{(\mathbf{Y} - \theta)^T P_i (\mathbf{Y} - \theta)}{\sigma^2}, \quad i = 1, 2.$$

Show that if $Q_i \sim \chi_{r_i}^2$ and $Q_1 - Q_2 \geq 0$, then $Q_1 - Q_2$ and Q_2 are independently distributed as $\chi_{r_1-r_2}^2$ and $\chi_{r_2}^2$, respectively.

Required theory (from the uploaded lecture notes)

Quadratic forms and Craig's theorem. If $Z \sim N(0, I)$ and P is symmetric idempotent of rank r , then $Z^T P Z \sim \chi_r^2$. For two such projection matrices A, B , Craig's theorem says

$$Z^T A Z \perp Z^T B Z \iff AB = 0.$$

Step 1: standardize. Set $Z = (Y - \theta)/\sigma \sim N_n(0, I)$. Then $Q_i = Z^T P_i Z$.

Step 2: interpret the non-negativity assumption. Because $Q_1 - Q_2 = Z^T (P_1 - P_2) Z \geq 0$ almost surely and the normal distribution has full support, $P_1 - P_2$ must be positive semidefinite. For orthogonal projections, this implies the range of P_2 is contained in the range of P_1 ; hence $P_2 P_1 = P_1 P_2 = P_2$. Therefore

$$P_1 - P_2$$

is itself a symmetric idempotent matrix, and

$$(P_1 - P_2)P_2 = 0.$$

Step 3: identify the two quadratic forms. We have

$$Q_1 - Q_2 = Z^T (P_1 - P_2) Z, \quad Q_2 = Z^T P_2 Z.$$

Their ranks are $r_1 - r_2$ and r_2 , respectively, since the subspaces are nested. By the quadratic-form chi-square theorem,

$$Q_1 - Q_2 \sim \chi_{r_1-r_2}^2, \quad Q_2 \sim \chi_{r_2}^2.$$

Step 4: prove independence. Since $(P_1 - P_2)P_2 = 0$, Craig's theorem gives independence.

Final answer

$$Q_1 - Q_2 \sim \chi_{r_1 - r_2}^2, \quad Q_2 \sim \chi_{r_2}^2, \quad Q_1 - Q_2 \perp Q_2.$$

1.22 MVN Q12: cycle differences and chi-square

Assignment question

Suppose $\mathbf{Y} \sim N_3(0, I)$. Show that

$$\frac{1}{3} [(Y_1 - Y_2)^2 + (Y_2 - Y_3)^2 + (Y_3 - Y_1)^2] \sim \chi_2^2.$$

Does some multiple of

$$(Y_1 - Y_2)^2 + \cdots + (Y_{n-1} - Y_n)^2 + (Y_n - Y_1)^2$$

have a chi-squared distribution for general n ?

Required theory (from the uploaded lecture notes)

If $Z \sim N(0, I)$ and A is symmetric idempotent of rank r , then $Z^T A Z \sim \chi_r^2$. Therefore the strategy is to express the sum as a quadratic form and examine its eigenvalues.

Step 1: write the quadratic form. Let

$$L = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{pmatrix}.$$

Then the sum of the three squared differences equals $Y^T L Y$. Therefore the statistic is $Y^T (L/3) Y$.

Step 2: inspect eigenvalues. The matrix L has eigenvalues 0, 3, 3. Hence $L/3$ has eigenvalues 0, 1, 1. It is symmetric and idempotent with rank 2.

Step 3: apply the theorem. Therefore

$$\frac{1}{3} [(Y_1 - Y_2)^2 + (Y_2 - Y_3)^2 + (Y_3 - Y_1)^2] \sim \chi_2^2.$$

General n case. For the cycle graph on n vertices, the Laplacian L_n has eigenvalues

$$\lambda_k = 2 - 2 \cos \left(\frac{2\pi k}{n} \right), \quad k = 0, \dots, n-1.$$

A scalar multiple cL_n is an idempotent projection only if all nonzero eigenvalues are equal to $1/c$. For $n = 3$, the two nonzero eigenvalues are both 3. For $n > 3$, the nonzero eigenvalues are not all equal. Thus there is no single scalar multiple that converts the full cycle quadratic form into an ordinary χ_{n-1}^2 for general $n > 3$.

Final answer

For $n = 3$, $c = 1/3$ works; for general $n > 3$, no scalar multiple gives a chi-square law.

For $n = 2$ the duplicated cycle expression is a special degenerate case and $c = 1/4$ gives χ_1^2 ; the substantive general case is $n \geq 3$.

2 Problems Set 2: Least Squares Regression

2.1 Model used throughout PS2

Assignment question

Throughout this problem set, consider the multiple linear regression model

$$\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

with $X = [\mathbf{1}, \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(p)}]$ and $\mathbb{E}(\boldsymbol{\varepsilon}) = 0$, $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 I_n$.

Required theory (from the uploaded lecture notes)

Least-squares theory from the lecture notes. The least-squares objective is

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - X\boldsymbol{\beta}\|^2.$$

When $X^T X$ is invertible,

$$\hat{\boldsymbol{\beta}}_{OLS} = (X^T X)^{-1} X^T \mathbf{Y}.$$

The projection matrix onto $C(X)$ is

$$P_X = X(X^T X)^{-1} X^T,$$

and the residual-maker is $M_X = I - P_X$. The residual is $\mathbf{e} = M_X \mathbf{Y}$, and

$$RSS = \mathbf{Y}^T M_X \mathbf{Y}.$$

Under only $\mathbb{E}\boldsymbol{\varepsilon} = 0$ and $\text{Var} \boldsymbol{\varepsilon} = \sigma^2 I$, $\hat{\boldsymbol{\beta}}_{OLS}$ is unbiased and

$$\text{Var}(\hat{\boldsymbol{\beta}}_{OLS}) = \sigma^2 (X^T X)^{-1}.$$

Under normality, $\hat{\boldsymbol{\beta}}_{OLS}$ and RSS are independent and $RSS/\sigma^2 \sim \chi_{n-p-1}^2$ for full column rank.

2.2 PS2 Q1: multiple versus simple regression

Assignment question

Consider the least square solution $\hat{\beta}_{OLS}$ from the multiple regression model. For the j -th simple regression

$$Y_i = \beta_0 + \beta_j x_{ij} + \varepsilon_i,$$

let $\hat{\beta}_{j,S}$ be the simple-regression slope and

$$\hat{\beta}_S = (\bar{Y}, \hat{\beta}_{1,S}, \dots, \hat{\beta}_{p,S})^T.$$

(a) Compare $\hat{\beta}_{OLS}$ and $\hat{\beta}_S$. (b) When will these two solutions be equal?

Required theory (from the uploaded lecture notes)

Adjusted versus marginal slopes. In the multiple regression, β_j measures the effect of x_j while holding the other regressors fixed. A simple-regression slope measures the marginal linear relationship between Y and x_j without adjustment for the other regressors. Thus they need not agree. With an intercept, it is convenient to center all columns first.

Step 1: center the variables. Let $\mathbf{x}_j$ be the centered j th regressor and Z contain the other centered regressors. The fitted multiple model is $\mathbf{y}_c = \mathbf{x}_j \hat{\beta}_{j,OLS} + Z\hat{\gamma} + \mathbf{e}$, with $\mathbf{x}_j^T \mathbf{e} = 0$.

Step 2: compute the simple slope from the multiple fit. The simple slope is

$$\hat{\beta}_{j,S} = \frac{\mathbf{x}_j^T \mathbf{y}_c}{\mathbf{x}_j^T \mathbf{x}_j}.$$

Substitute the multiple fitted decomposition:

$$\begin{aligned} \hat{\beta}_{j,S} &= \frac{\mathbf{x}_j^T (\mathbf{x}_j \hat{\beta}_{j,OLS} + Z\hat{\gamma} + \mathbf{e})}{\mathbf{x}_j^T \mathbf{x}_j} \\ &= \hat{\beta}_{j,OLS} + \frac{\mathbf{x}_j^T Z\hat{\gamma}}{\mathbf{x}_j^T \mathbf{x}_j}. \end{aligned}$$

Part (a): comparison. The intercept in the simple regression is always $\bar{Y}$, while each simple slope generally differs from the corresponding multiple-regression coefficient by the contribution of the other fitted regressors.

Part (b): equality condition. Therefore, for each j ,

$$\hat{\beta}_{j,S} = \hat{\beta}_{j,OLS} \iff \mathbf{x}_j^T Z\hat{\gamma} = 0.$$

A strong design-based sufficient condition is $\mathbf{x}_j^T \mathbf{x}_k = 0$ for every $k \neq j$, i.e. the centered regressors are mutually orthogonal. Another special-data possibility is a cancellation among the other fitted coefficients.

Final answer

$$\hat{\beta}_{j,S} = \hat{\beta}_{j,OLS} + \frac{\mathbf{x}_j^T Z\hat{\gamma}}{\mathbf{x}_j^T \mathbf{x}_j}.$$

They are automatically equal when the j th centered regressor is orthogonal to every other centered regressor.

2.3 PS2 Q2: orthogonal design

Assignment question

Assume

$$\mathbf{1}^T \mathbf{x}^{(j)} = 0, \quad (\mathbf{x}^{(k)})^T \mathbf{x}^{(j)} = 0 \quad (k \neq j).$$

(a) Find $\hat{\beta}_{OLS}$. (b) Find the RSS.

Required theory (from the uploaded lecture notes)

Under the stated orthogonality, the columns of X are mutually orthogonal. Therefore $X^T X$ is diagonal with first entry n and remaining entries $\|\mathbf{x}^{(j)}\|^2$. Hence $(X^T X)^{-1}$ is also diagonal. The OLS formula then separates into one scalar formula per coefficient.

Step 1: compute $X^T X$. The intercept column is orthogonal to each centered regressor, so

$$X^T X = \text{diag} \left(n, \|x^{(1)}\|^2, \dots, \|x^{(p)}\|^2 \right).$$

Step 2: multiply by $X^T Y$. Since $\mathbf{1}^T \mathbf{Y} = n\bar{Y}$,

$$\hat{\beta}_0 = \bar{Y}, \quad \hat{\beta}_j = \frac{(x^{(j)})^T \mathbf{Y}}{\|x^{(j)}\|^2}, \quad j = 1, \dots, p.$$

Step 3: compute RSS. Use $RSS = Y^T Y - Y^T X (X^T X)^{-1} X^T Y$. Therefore

$$RSS = \mathbf{Y}^T \mathbf{Y} - n\bar{Y}^2 - \sum_{j=1}^p \frac{((x^{(j)})^T \mathbf{Y})^2}{\|x^{(j)}\|^2}.$$

Equivalently, with $\mathbf{Y}_c = \mathbf{Y} - \bar{Y}\mathbf{1}$,

$$RSS = \|\mathbf{Y}_c\|^2 - \sum_{j=1}^p \frac{((x^{(j)})^T \mathbf{Y}_c)^2}{\|x^{(j)}\|^2}.$$

Final answer

Orthogonality turns the matrix normal equations into independent one-variable equations and makes the RSS decomposition additive across regressors.

2.4 PS2 Q3: why orthogonal covariates minimize total coefficient variance

Assignment question

Assume

$$\mathbf{1}^T \mathbf{x}^{(j)} = 0, \quad \|\mathbf{x}^{(j)}\| = c,$$

for all $j = 1, \dots, p$. Show that $\sum_{j=1}^p \text{Var}(\hat{\beta}_{j,OLS})$ is minimized when the covariates are orthogonal.

Required theory (from the uploaded lecture notes)

For full rank OLS,

$$\text{Var}(\hat{\beta}_{OLS}) = \sigma^2 (X^T X)^{-1}.$$

Because the intercept is orthogonal to the regressors, the slope block is G^{-1} where G is the Gram matrix of the centered covariates. The diagonal of G is fixed at c^2 . The trace of the inverse is minimized when all eigenvalues of G are equal. Equality of all eigenvalues, given the fixed diagonal sum, occurs for $G = c^2 I$, exactly the orthogonal-design case.

Step 1: isolate the slope block. Write

$$X^T X = \begin{pmatrix} n & 0 \\ 0 & G \end{pmatrix},$$

where $G_{jk} = (x^{(j)})^T x^{(k)}$ and $G_{jj} = c^2$. Thus

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) = \sigma^2 \text{tr}(G^{-1}).$$

Step 2: use the eigenvalues. Let $\lambda_1, \dots, \lambda_p > 0$ be the eigenvalues of G . Then

$$\sum_i \lambda_i = \text{tr}(G) = pc^2.$$

By Cauchy-Schwarz,

$$\left(\sum_i \lambda_i \right) \left(\sum_i \frac{1}{\lambda_i} \right) \geq p^2,$$

so

$$\text{tr}(G^{-1}) \geq \frac{p^2}{pc^2} = \frac{p}{c}.$$

Equality holds iff $\lambda_1 = \dots = \lambda_p = c^2$.

Step 3: interpret equality. A Gram matrix with diagonal entries all c^2 and all eigenvalues equal to c^2 must be $G = c^2 I$. Thus all off-diagonal inner products are zero, i.e. the covariates are orthogonal.

Final answer

$$\sum_{j=1}^p \text{Var}(\hat{\beta}_j) \geq \sigma^2 \frac{p}{c^2},$$

with equality exactly for the orthogonal design $G = c^2 I$.

2.5 PS2 Q4: BLUE for $c^T \beta$

Assignment question

Let X have rank $p + 1$. Show that, for any parameter of the form $c^T \beta$, the estimator $c^T \hat{\beta}_{OLS}$ is the best linear unbiased estimator.

Required theory (from the uploaded lecture notes)

Gauss–Markov/BLUE result. With X full column rank and $\text{Var}(\varepsilon) = \sigma^2 I$, $\hat{\beta}_{OLS}$ is unbiased and has covariance $\sigma^2(X^T X)^{-1}$. Any scalar $c^T \hat{\beta}_{OLS}$ is therefore a linear unbiased estimator. To prove it is BLUE, compare it with any other linear unbiased estimator $a^T Y$ and show that the variance difference is a nonnegative quadratic norm.

Step 1: unbiasedness. Since $\mathbb{E}Y = X\beta$,

$$\mathbb{E}(c^T \hat{\beta}) = c^T (X^T X)^{-1} X^T X \beta = c^T \beta.$$

Thus it is unbiased.

Step 2: take an arbitrary competing LUE. Let $a^T Y$ be another linear unbiased estimator of $c^T \beta$. Unbiasedness for every β requires

$$a^T X = c^T.$$

Set $b = X(X^T X)^{-1}c$, so $c = X^T b$ and $P_X b = b$. The orthogonal decomposition of a into $P_X a$ and $M_X a$ gives

$$a^T a = b^T b + \|M_X a\|^2.$$

More explicitly, the difference in variances can be written as

$$\text{Var}(a^T Y) - \text{Var}(c^T \hat{\beta}_{OLS}) = \sigma^2 \|M_X a\|^2 \geq 0.$$

Step 3: conclude optimality. The variance is never smaller than that of $c^T \hat{\beta}_{OLS}$. Equality occurs only when $M_X a = 0$, i.e. the competitor is itself the OLS linear estimator.

Final answer

$$c^T \hat{\beta}_{OLS} \text{ is the unique BLUE of } c^T \beta.$$

2.6 PS2 Q5: variance of $\hat{\beta}_1$ with two centered regressors

Assignment question

Let

$$Y_i = \beta_0 + \beta_1(x_{i1} - \bar{x}_1) + \beta_2(x_{i2} - \bar{x}_2) + \varepsilon_i,$$

with $\mathbb{E}\varepsilon = 0$, $\text{Var}\varepsilon = \sigma^2 I$. If $\hat{\beta}_1$ is the least-squares estimate of β_1 , show that

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i (x_{i1} - \bar{x}_1)^2 (1 - r^2)},$$

where r is the correlation coefficient of the pairs (x_{i1}, x_{i2}) .

Required theory (from the uploaded lecture notes)

For a centered two-predictor model, the intercept column is orthogonal to both centered regressors. Hence the slope covariance matrix is $\sigma^2 G^{-1}$, where

$$G = \begin{pmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{pmatrix}, \quad S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2, \quad r = \frac{S_{12}}{\sqrt{S_{11}S_{22}}}.$$

The inverse of a 2×2 matrix gives the required expression.

Step 1: form the Gram matrix. We have

$$G = \begin{pmatrix} S_{11} & S_{12} \\ S_{12} & S_{22} \end{pmatrix}.$$

Step 2: invert it. Since

$$G^{-1} = \frac{1}{S_{11}S_{22} - S_{12}^2} \begin{pmatrix} S_{22} & -S_{12} \\ -S_{12} & S_{11} \end{pmatrix},$$

the (1,1) entry is $S_{22}/(S_{11}S_{22} - S_{12}^2)$. Thus

$$\text{Var}(\hat{\beta}_1) = \sigma^2 \frac{S_{22}}{S_{11}S_{22} - S_{12}^2}.$$

Step 3: introduce the correlation coefficient. Because $r^2 = S_{12}^2/(S_{11}S_{22})$,

$$S_{11}S_{22} - S_{12}^2 = S_{11}S_{22}(1 - r^2).$$

Therefore

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{11}(1 - r^2)} = \frac{\sigma^2}{\sum_i (x_{i1} - \bar{x}_1)^2 (1 - r^2)}.$$

Why this step leads to the next one

As $|r|$ approaches 1, the predictors become nearly collinear, $1 - r^2$ becomes small, and the variance of the estimated partial slope explodes. This is the algebraic form of multicollinearity.

2.7 PS2 Q6: minimum-variance unbiased quadratic estimator of error variance

Assignment question

Let $Y_1, \dots, Y_n$ be independent with common variance σ^2 and common third and fourth moments μ_3, μ_4 about their means. If $\mathbb{E}[Y] = X\beta$, where X is an $n \times p$ matrix of rank p , then $(n - p)S^2$ is the unique nonnegative quadratic unbiased estimate of $(n - p)\sigma^2$ with minimum variance when $\mu_4 = 3\sigma^4$ or when the diagonal elements of P are all equal. [Hint: Seber and Lee, Chapter 3, Section 3.3.]

Required theory (from the uploaded lecture notes)

Quadratic-estimator theory. Let P be the orthogonal projection onto $C(X)$ and $M = I - P$. Then $\text{rank}(M) = n - p$, $MX = 0$, and the residual sum of squares is

$$RSS = (Y - X\hat{\beta})^T(Y - X\hat{\beta}) = Y^TMY.$$

Thus $S^2 = RSS/(n - p)$. For independent components with common second and fourth moments, the variance of a symmetric quadratic form Y^TAY contains the two terms

$$2\sigma^4 \text{tr}(A^2) + (\mu_4 - 3\sigma^4) \sum_i a_{ii}^2.$$

Unbiasedness for σ^2 requires $AX = 0$ and $\text{tr}(A) = n - p$ (after scaling), while nonnegativity means $A \succeq 0$.

Step 1: identify the residual projection. Take $M = I - P$. Then M is symmetric idempotent, $MX = 0$, and $\text{tr}(M) = \text{rank}(M) = n - p$. Hence

$$\mathbb{E}(Y^T MY) = \sigma^2 \text{tr}(M) = (n - p)\sigma^2,$$

so $RSS = Y^T MY$ is unbiased for $(n - p)\sigma^2$, and therefore

$$S^2 = \frac{RSS}{n - p}$$

is unbiased for σ^2 .

Step 2: why M is optimal when $\mu_4 = 3\sigma^4$. For a competing nonnegative unbiased quadratic estimator $Y^T AY$, $A \succeq 0$, $AX = 0$, and $\text{tr} A = n - p$. Since A acts on an $(n - p)$ -dimensional residual space, let its nonzero eigenvalues be $\lambda_1, \dots, \lambda_{n-p}$. Then

$$\text{tr}(A^2) = \sum_j \lambda_j^2 \geq \frac{(\sum_j \lambda_j)^2}{n - p} = n - p,$$

with equality iff all nonzero eigenvalues are 1. That equality condition makes A exactly the projection M . When $\mu_4 = 3\sigma^4$, the fourth-moment correction vanishes, so minimizing $\text{tr}(A^2)$ gives the unique minimum-variance estimator M .

Step 3: equal diagonal condition. If the diagonal elements of P are all equal, then the diagonal elements of $M = I - P$ are also all equal. Since $\text{tr} A = n - p$, Cauchy–Schwarz implies

$$\sum_i a_{ii}^2 \geq \frac{(n - p)^2}{n},$$

with equality at the equal-diagonal projection M . Together with the preceding eigenvalue argument, this gives the stated minimum-variance result under the second condition in the problem statement.

Step 4: identify the estimator. Because $Y^T MY = RSS$ and $S^2 = RSS/(n - p)$, the unique estimator is the usual residual variance estimator.

Final answer

$$(n - p)S^2 = RSS = Y^T(I - P)Y, \quad S^2 = \frac{RSS}{n - p}.$$

It is unbiased and is the unique nonnegative quadratic unbiased estimator with minimum variance under the conditions stated in the problem.

2.8 PS2 B1: independence of OLS and RSS under normality

Assignment question

Suppose, in addition to the regression model, that the errors are normal as stated in the assignment. Show that $\hat{\beta}_{OLS}$ and RSS are independent.

Required theory (from the uploaded lecture notes)

Under normal errors, $Y \sim N(X\beta, \sigma^2 I)$. Both $\hat{\beta}_{OLS}$ and the residual vector are linear functions of Y , so they are jointly normal. For jointly normal quantities, zero covariance implies independence. Use

$$\hat{\beta} = (X^T X)^{-1} X^T Y, \quad e = M_X Y,$$

with $M_X X = 0$ and $P_X X = X$.

Step 1: compute the cross-covariance. Since $\text{Var}(Y) = \sigma^2 I$,

$$\begin{aligned}\text{Cov}(\hat{\beta}, e) &= (X^T X)^{-1} X^T \text{Var}(Y) M_X \\ &= \sigma^2 (X^T X)^{-1} X^T M_X.\end{aligned}$$

But $X^T M_X = 0$ because M_X projects onto the orthogonal complement of $C(X)$. Hence $\text{Cov}(\hat{\beta}, e) = 0$.

Step 2: replace the residual vector by RSS. RSS is a function of e , namely $RSS = e^T e$. Since $\hat{\beta}$ is independent of the entire residual vector e , it is independent of any function of e , including RSS.

Final answer

$$\hat{\beta}_{OLS} \perp RSS.$$

2.9 PS2 B2: consistency of MLEs

Assignment question

Suppose in addition to the regression model that ε_i are i.i.d. normal as stated in the assignment. Show that both $\hat{\beta}_{ML}$ and $\hat{\sigma}_{ML}^2$ are consistent estimators.

Required theory (from the uploaded lecture notes)

Normal-regression MLE theory. Under Gaussian errors,

$$\ell(\beta, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (Y - X\beta)^T (Y - X\beta).$$

Maximizing over β gives the OLS estimator

$$\hat{\beta}_{ML} = \hat{\beta}_{OLS},$$

and maximizing over σ^2 gives

$$\hat{\sigma}_{ML}^2 = \frac{RSS}{n}.$$

For consistency of $\hat{\beta}$, assume the standard fixed-design condition that $(X^T X)/n \rightarrow Q$ with Q positive definite. Then $(X^T X)^{-1} = O(n^{-1})$ and the estimation error vanishes in probability. For the variance estimator, under normality $RSS/\sigma^2 \sim \chi_{n-p-1}^2$ for full rank.

Step 1: derive the MLEs. The likelihood is maximized over β exactly when the residual sum of squares is minimized, hence

$$\hat{\beta}_{ML} = \hat{\beta}_{OLS}.$$

Then the profile likelihood in σ^2 gives

$$\hat{\sigma}_{ML}^2 = \frac{RSS}{n}.$$

Step 2: prove $\hat{\beta}_{ML}$ is consistent. Write

$$\hat{\beta}_{OLS} - \beta = (X^T X)^{-1} X^T \varepsilon.$$

Its covariance is

$$\text{Var}(\hat{\beta}_{OLS}) = \sigma^2 (X^T X)^{-1}.$$

Under $(X^T X)/n \rightarrow Q > 0$, this covariance tends to 0. Since the estimator is unbiased, Chebyshev's inequality gives

$$\hat{\beta}_{OLS} \xrightarrow{p} \beta.$$

Step 3: prove $\hat{\sigma}_{ML}^2$ is consistent. For fixed rank $p + 1$, under normality

$$\frac{RSS}{\sigma^2} \sim \chi_{n-p-1}^2.$$

Therefore

$$\mathbb{E} \left[\frac{RSS}{n} \right] = \sigma^2 \frac{n-p-1}{n} \rightarrow \sigma^2,$$

and

$$\text{Var} \left(\frac{RSS}{n} \right) = \frac{2\sigma^4(n-p-1)}{n^2} \rightarrow 0.$$

Thus

$$\hat{\sigma}_{ML}^2 = \frac{RSS}{n} \xrightarrow{p} \sigma^2.$$

Literal wording of the uploaded assignment. The PDF prints $\varepsilon_i \stackrel{iid}{\sim} N(0, 1)$ in this question. If that is interpreted literally, $\sigma^2 = 1$ is not an unknown parameter, so there is no variance parameter to estimate. The intended statistical statement is the standard $N(0, \sigma^2)$ model; under that interpretation the consistency calculation above applies. In the literal $\sigma^2 = 1$ case, $RSS/n \xrightarrow{p} 1$.

Final answer

$$\boxed{\hat{\beta}_{ML} = \hat{\beta}_{OLS} \xrightarrow{p} \beta, \quad \hat{\sigma}_{ML}^2 = RSS/n \xrightarrow{p} \sigma^2.}$$